

FALKO DUWANOFF

Software Engineer · Machine Learning · C++ · Python · Rust · Embedded & Android

Germany · Open to Remote / International Opportunities

- f.duwanoff@gmail.com
 - www.linkedin.com/in/falko-duwanoff-a661a9255
 - <https://fuzzyart.github.io/dev-portfolio/>
-

PROFILE

Software Engineer and Machine Learning specialist with several years of hands-on experience developing ML and software solutions for **Linux, Android, embedded systems, GPUs, and spectral imaging**.

Experienced in implementing machine-learning algorithms from scratch, developing native C/C++ software, integrating native libraries into Android through JNI, and building end-to-end ML workflows from data acquisition to model generation and classification.

Currently focused on independent projects involving **Rust, machine learning, Linux, CI/CD, and developer tooling**.

PROFESSIONAL EXPERIENCE

Fraunhofer IFF – Research Assistant
2015 – 2021

Developed and evaluated machine-learning and software solutions for mobile, desktop, GPU, and embedded platforms.

- Implemented **MLP and Growing Neural Gas (GNG) pipelines from scratch**, including interfaces compatible with MATLAB workflows.
- Developed custom **clustering algorithms** broadly inspired by the Growing Neural Gas approach.
- Worked with **hyperspectral and spectral imaging** for automated analysis and classification.
- Developed and maintained **Android applications** using Java and the Android SDK; independently implemented most of a smartphone-based spectral-imaging application.
- Built an end-to-end workflow in which smartphone-acquired image data was uploaded to a server, processed for **ML model generation**, and subsequently used for classification of newly captured images.
- Extended an existing Android application with new and enhanced features for processing **spectral measurements** from spectrometer systems.
- Developed native C/C++ components and integrated them into Android applications using **Java JNI**.

- Used **OpenCV** for image processing, **OpenMP** for CPU parallelization, and **OpenCL** for GPU-based computation.
- Developed and evaluated software across **Linux PCs, GPUs, embedded Linux boards, and smartphones**.
- Evaluated algorithm performance and practical applicability across different hardware platforms.

Nexus Marabu — Java Backend Developer

2023

- Developed a **Java backend interface** for a server application used in a medical environment.
- Worked on backend software development and system integration.

Independent Software & Machine Learning Projects

2023 – Present

Developing and maintaining independent software projects with a focus on machine learning, systems programming, Linux, and developer tooling.

neurogas — Rust Growing Neural Gas

- Designed and implemented a **Growing Neural Gas algorithm as a reusable Rust crate**.
- Focused on clean library design and practical implementation of an ML algorithm in Rust.

Machine Learning Pipelines

- Develop and experiment with ML workflows using **Python, TensorFlow, PyTorch, NumPy, Pandas, Matplotlib, and scikit-learn**.
- Implement and evaluate ML approaches with a focus on reproducible experimentation.

CLI Backup Tool

- Developing a Linux command-line backup application with a focus on reliable file handling and systems programming.

Linux CI/CD

- Build and maintain automated build and test workflows using **GitLab CI/CD**.
- Maintain Linux-based development and infrastructure environments, including **Proxmox**.

EDUCATION

Diploma in Technical Computer Science (Dipl.-Ing.)

Otto-von-Guericke Universität · Magdeburg · Germany

PROFESSIONAL TRAINING

Expert for Information Security and Business Continuity

de-dsb GmbH · 2024

BSI IT-Grundschutz · Business Continuity Management · ISO/IEC 27001

TECHNICAL SKILLS

Languages C · C++ · Rust · Python · Java · SQL

Machine Learning & Data Science TensorFlow · PyTorch · scikit-learn · NumPy · Pandas · Matplotlib · MLP · Growing Neural Gas · Clustering · Image Classification

Computer Vision & Spectral Imaging OpenCV · Image Processing · Hyperspectral Imaging · Spectral Data Analysis

Performance & Systems OpenCL · OpenMP · GPU Computing · Linux · Embedded Linux · Embedded Systems · IoT

Mobile & Native Development Android SDK · Java · JNI · Native C/C++

Development & Infrastructure Git · GitLab · GitLab CI/CD · Proxmox · Automated Build & Test

Databases SQL · NoSQL

Information Security BSI IT-Grundschutz · ISO/IEC 27001 · Business Continuity Management

LANGUAGES

German — Native **English** — Professional Working Proficiency